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## REVERSE FLOW GAS TURBINE ENGINE HAVING ELECTRIC MACHINE

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### Abstract

An aircraft engine assembly includes a gas turbine engine having an intake channel configured to receive an incoming flow of air and form an intake flow of air, the intake channel configured to turn the received incoming flow of air from an incoming flow direction to a first axial direction of the gas turbine engine, the incoming flow direction reverse of the first axial direction, and an electric machine coupled with the low pressure shaft and located at the aft end of the gas turbine engine proximate the intake channel, the electric machine in heat exchange communication with the intake flow of air such that the electric machine transfers heat to the incoming flow of air within the intake channel when the electric machine is operated.

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## Background/Summary

**PRIORITY INFORMATION [0001]** The present application claims priority to U.S. patent application Ser. No. 18/990,417, filed on Dec. 20, 2024, which is a Continuation in-part of U.S. patent application Ser. No. 18/307,938, filed on Apr. 27, 2023, which claims priority to Polish Patent Application Number P.443814, filed on Feb. 17, 2023. U.S. patent application Ser. No. 18/990,417, U.S. patent application Ser. No. 18/307,938 and Polish Patent Application Number P.443814 are hereby incorporated by reference in their entirety for all purposes.

### FIELD

[0002] The present disclosure relates generally to a reverse flow gas turbine engine having an electric machine.

### BACKGROUND

[0003] A gas turbine engine generally includes a turbomachine and a rotor assembly. Gas turbine engines, such as turboprop engines, may be used for aircraft propulsion. In the case of a turboprop engine, the rotor assembly may be configured as a variable pitch propeller. In some installations the gas turbine engine is oriented in a reverse flow configuration such that an air flow provided by forward motion of an aircraft is received by the gas turbine engine and turned to flow in a reverse direction through the turbomachinery of the gas turbine engine before an exhaust is discharged from the engine. Locating auxiliary components in such a reverse flow configuration remains an area of interest.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0004] A full and enabling disclosure of the presently described technology, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0005] **FIG. 1** is a cross-sectional view of a gas turbine engine in accordance with an exemplary aspect of the present disclosure.

[0006] **FIG. 2** is a schematic view of a gas turbine engine with an electric machine in accordance with the present disclosure.

[0007] **FIG. 3** is a schematic view of a gas turbine engine with an electric machine in accordance with the present disclosure.

[0008] **FIG. 4** is a schematic view of a gas turbine engine with an electric machine in accordance with the present disclosure.

[0009] **FIG. 5** is a schematic view of a gas turbine engine with an electric machine in accordance

with the present disclosure.

[0010] FIG. **6** is a view of a controller in accordance with the present disclosure.

[0011] FIG. **7A** is a cross sectional view of a non-annular portion of a flow path of an intake channel in accordance with the present disclosure.

[0012] FIG. **7B** is a cross sectional view of an annular portion of a flow path of an intake channel in accordance with the present disclosure.

[0013] FIG. **8** is a flow chart describing a method of operating a gas turbine engine having an electric machine in accordance with the present disclosure.

[0014] FIG. **9** is a cross-sectional view of an example sealing configuration that can be implemented in examples disclosed herein.

[0015] FIG. **10** illustrates an isometric view of an example strut fairing that can be implemented in examples disclosed herein.

[0016] FIGS. **11** and **12** are cross-sectional views of the example fairing of FIG. **10**.

[0017] FIG. **13** illustrates an example airfoil that can be implemented in examples disclosed herein.

[0018] FIGS. **14** and **15** illustrate example sealing configurations that can be implemented in examples disclosed herein.

[0019] FIG. **16** illustrates an example shroud hanger sealing configuration in an example engine casing assembly that can be implemented in examples disclosed herein.

[0020] FIG. **17** illustrates another example sealing configuration that can be implemented in examples disclosed herein.

[0021] FIG. **18** is a cross-sectional view of an example engine casing that can be implemented in examples disclosed herein.

[0022] FIGS. **19-21B** are top views of the example airfoil section of FIG. **18**.

#### DETAILED DESCRIPTION

[0023] Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

[0024] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

[0025] The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

[0026] The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C.

[0027] The term “gas turbine engine” refers to an engine having a turbomachine as all or a portion of its power source.

[0028] The term “combustion section” refers to any heat addition system for a turbomachine. For example, the term combustion section may refer to a section including one or more of a deflagrative combustion assembly, a rotating detonation combustion assembly, a pulse detonation combustion assembly, or other appropriate heat addition assembly. In certain example embodiments, the combustion section may include an annular combustor, a can combustor, a cannular combustor, a trapped vortex combustor (TVC), or other appropriate combustion system, or combinations thereof.

[0029] The terms “low” and “high”, or their respective comparative degrees (e.g., -er, where applicable), when used with a compressor, a turbine, a shaft, or spool components, etc. each refer to relative pressure within an engine unless otherwise specified. For example, a “low turbine” or “low pressure turbine” defines a component configured to operate at a pressure lower than a “high

pressure turbine” of the engine.

[0030] The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of a vehicle such as an aircraft. For example, with regard to an aircraft, forward refers to a position closer to a nose of the aircraft and aft refers to a position closer to an empennage of the aircraft.

[0031] The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

[0032] As used herein, the terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to a centerline of the gas turbine engine. Moreover, the terms “radial” and “radially” refer to directions and orientations that extend substantially perpendicular to the centerline of the gas turbine engine. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the centerline of the gas turbine engine.

[0033] As used herein, the terms “system,” “unit,” “module,” “engine,” “component,” etc., may include a hardware and/or software system that operates to perform one or more functions. For example, a module, unit, or system may include a computer processor, controller, and/or other logic-based device that performs operations based on instructions stored on a tangible and non-transitory computer readable storage medium, such as a computer memory. Alternatively, a module, unit, or system may include a hard-wired device that performs operations based on hard-wired logic of the device. Various modules, units, engines, and/or systems shown in the attached figures may represent the hardware that operates based on software or hardwired instructions, the software that directs hardware to perform the operations, or a combination thereof.

[0034] As will be discussed in more detail below, the subject matter of the present disclosure is directed generally to locating an electric machine near an intake channel of a reverse flow turboprop engine at a location which is closer to an aft end of the turboprop engine than to a forward end of the turboprop engine. The electric machine is rotatably coupled to a low pressure shaft of the reverse flow turboprop engine and as a consequence of the reverse flow configuration the low pressure shaft extends aft of a core of the turboprop engine. The electric machine can be operated as a generator and/or motor for use in either adding power to and/or extracting power from the low pressure shaft. Placement of the electric machine in the proximate location described above permits an exchange of heat between the electric machine and a flow of air traversing through an intake channel of the engine. Such exchange of heat can provide tighter packaging of the electric machine and/or higher heat generating operating demands placed upon the electric machine.

[0035] To accommodate the placement of the electric machine in an aft location in at least one embodiment, an intake channel may be provided that forms a non-annular flow path at an inlet to the intake channel which then changes to an annular flow path around the LP shaft prior to air being delivered to a compressor of the gas turbine engine.

[0036] Referring now to the drawings, wherein identical numerals indicate the same elements throughout the figures, FIG. 1 is a schematic cross-sectional view of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure. More particularly, for the embodiment of FIG. 1, the gas turbine engine is a reverse flow turboprop engine 10, referred to herein as “turboprop engine 10.” As shown in FIG. 1, turboprop engine 10 defines an axial direction A (extending parallel to a longitudinal centerline or central axis 12 provided for reference), a radial direction R, and a circumferential direction C (not shown) disposed about the axial direction A. Turboprop engine 10 generally includes a propeller section 14 and a core turbine engine 16 disposed aft of the propeller section 14 from an aircraft perspective, the propeller section 14 being operable with, and driven by, core turbine engine 16.

[0037] The exemplary core turbine engine 16 depicted generally includes a substantially tubular

outer casing **18** extending generally along axial direction A. Outer casing **18** generally encloses core turbine engine **16** and may be formed from a single casing or multiple casings. Core turbine engine **16** includes, in a serial flow relationship, a compressor **22**, a combustion section **26**, a high pressure (HP) turbine **28**, a low pressure (LP) turbine **30**, and an exhaust section **32**. An air flow path generally extends through compressor **22**, combustion section **26**, HP turbine **28**, LP turbine **30**, and exhaust section **32** which are in fluid communication with each other.

[0038] An HP shaft or spool **34** drivingly connects the HP turbine **28** to the compressor **22**. An LP shaft or spool **36** drivingly connects the LP turbine **30** to propeller section **14** of the turboprop engine **10**. For the embodiment depicted, propeller section **14** includes a variable pitch propeller **38** having a plurality of propeller blades **40** coupled to a disk **42** in a spaced apart manner. As depicted, the propeller blades **40** extend outwardly from disk **42** generally along the radial direction R. Each propeller blade **40** is rotatable relative to the disk **42** about a pitch axis P by virtue of the propeller blades **40** being operatively coupled to a suitable actuation member **44** configured to collectively vary the pitch of the propeller blades **40** in unison. The propeller blades **40**, disk **42**, and actuation member **44** are together rotatable about the longitudinal centerline **12** by LP shaft **36** across a power gear box **46**. The power gear box **46** includes a plurality of gears for stepping down the rotational speed of the LP shaft **36** to a more efficient rotational fan speed and is attached to one or both of a core frame or a fan frame through one or more coupling systems. Disk **42** is covered by a rotatable front hub **48** aerodynamically contoured to promote an airflow through the plurality of propeller blades **40**.

[0039] During operation of the turboprop engine **10**, a volume of air **50** (also referred to as a free stream flow of air **51** prior to its encounter with the propeller **38**, and referred to as an incoming flow of air **50** after passage through the propeller **38**) passes through blades **40** of propeller **38** and is urged toward a radial inlet **52** of core turbine engine **16**. More specifically, turboprop engine **10** includes an intake channel **54** that defines radial inlet **52** that routes an inlet portion of air **53** of the flow of air **50** from inlet **52** downstream to compressor **22**. Though the inlet **52** is depicted as a radial inlet in the embodiment of FIG. 1, other configurations of inlet **52** are also contemplated. For example, the inlet **52** can also take the form of an inlet arranged in an axial direction to capture the inlet portion of air **53** of the volume of air **50**. The inlet portion of air **53** of the flow of air **50** captured by the inlet **52** is referred to herein as an intake flow of air. The intake channel **54** defines the intake flow of air and generally extends from an inlet of the intake channel **54** to just upstream of the compressor **22**.

[0040] The turboprop engine **10** embodiments described herein are configured as reverse flow engines. Such engines are characterized by a general relationship between the direction of the flow of incoming air **50** (such direction can be used to characterize the relative motion of air during a mode of operation of the engine **10** such as a forward thrust mode) and that of the flow of air axially through the turboprop engine **10**. The flow of air through the core turbine engine **16** is generally reverse to that of the flow of incoming air **50**. Turning the flow from the direction of the incoming flow of air **50** to the axial direction through the core turbine engine **16** is usually performed by the intake channel **54**. The change of direction is reversed in that the bulk direction of the flow of air **50** (itself having a circumferential swirl component imparted by the propeller blades **40** in addition to a longitudinal component) is opposite, or reverse, to the bulk direction of air flow axially through the core turbine engine **16** (which itself also includes a longitudinal component but also include radial and circumferential components owing to the shape of the flow path and swirl induced by rotating turbomachinery components) during one or more phases of operation of the core turbine engine **16**. Thus, it will also be appreciated that the term “reverse” is a relative comparison of the longitudinal components of the bulk flow of air **50** and bulk flow of air axially within the engine **10**. Though the longitudinal direction of the flow of air **50** may not be perfectly parallel with the axial flow of air through the engine **10**, it will be appreciated that the longitudinal components of the directions the flow of air **50** and the axial flow are reversed.

[0041] Compressor **22** includes one or more sequential stages of compressor stator vanes **60**, one or more sequential stages of compressor rotor blades **62**, and an impeller **64**. Though the illustrated embodiment includes both axial and centrifugal flow compressors, in some forms the turboprop engine **10** can include just an axial flow compressor(s) or centrifugal flow compressor(s). The one or more sequential stages of compressor stator vanes **60** are coupled to the outer casing **18** and compressor rotor blades **62** are coupled to HP shaft **34** to progressively compress the air **53**. Impeller **64** further compresses air **53** and directs the compressed air **53** into combustion section **26** where air **53** mixes with fuel. Combustion section **26** includes a combustor **66** which combusts the air/fuel mixture to provide combustion gases **68**.

[0042] Combustion gases **68** flow through HP turbine **28** which includes one or more sequential stages of turbine stator vanes **70** and one or more sequential stages of turbine blades **72**. The one or more sequential stages of turbine stator vanes **70** are coupled to the outer casing **18** and turbine blades **72** are coupled to HP shaft **34** extract thermal and/or kinetic energy therefrom. Combustion gases **68** subsequently flow through LP turbine **30**, where an additional amount of energy is extracted through additional stages of turbine stator vanes **70** and turbine blades **72** coupled to LP shaft **36**. The energy extraction from HP turbine **28** supports operation of compressor **22** through HP shaft **34** and the energy extraction from LP turbine **30** supports operation of propeller section **14** through LP shaft **36**. Combustion gases **68** exit turboprop engine **10** through exhaust section **32**.

[0043] It will be understood that one or more rows of stator vanes **60** and **70** can be variable vanes controlled by a controller (see below with respect to controller **100**) in one form. Furthermore, with particular respect to stator vanes **70**, one or more rows of the stator vanes **70** can be variable.

[0044] In other exemplary embodiments, the turbine engine may include any suitable number of compressors, turbines, shafts, etc. For example, as will be appreciated. HP shaft **34** and LP shaft **36** may further be coupled to any suitable device for any suitable purpose. For example, in certain exemplary embodiments. turboprop engine **10** of FIG. **1** may be utilized in aeroderivative applications. Additionally, in other exemplary embodiments, turboprop engine **10** may include any other suitable type of combustor, and may not include the exemplary reverse flow combustor depicted.

[0045] The embodiment of turboprop engine **10** illustrated in FIG. **1** includes an electric machine **74** located aft of the core turbine engine **16** and rotatably coupled to the LP shaft **36**. In some forms the electric machine **74** is contained in an environmentally sealed housing which can be pressurized to minimize electrical corona and discharge effects. Further, in some forms the LP shaft **36** and electric machine **74** can be configured to rotate at a constant speed from idle to max power, with thrust of the engine **10** controlled by the variable pitch propeller **38**.

[0046] Given the coaxial relationship of the LP shaft **36** with the HP shaft **34**, in such an embodiment the LP shaft **36** is configured to extend aft of the core turbine engine **16**, and also further aft than the HP shaft **34**, despite the HP compressor **22** being the upstream-most compressor of the turboprop engine **10** (i.e., despite there being no low-pressure compressor upstream of the HP compressor **22** and downstream of the inlet **52**).

[0047] The electric machine **74** can be used in many different power configurations. In one form the electric machine is configured to extract power from the LP shaft **36** when the machine **74** operates as a generator. The extraction of mechanical power from the LP shaft **36** and conversion to electric power can be used to charge an on-board power storage device such as a battery, or alternatively to provide power to another electrical device (e.g., an electric motor, an electrical accessory on an aircraft, etc.). In other forms, the electric machine **74** can be used as a motor to provide power to the LP shaft **36** to supplement power extracted by the LP turbine **30** from the combustion gases **68**. In these forms, the electric machine **74** can be configured to provide a minimum of 10% of supplemental thrust to the engine **10**, a minimum of 20% of supplemental thrust to the engine **10**, and up to 40% of supplemental thrust to the engine **10** in various embodiments. In still other forms, the electric machine **74** can be configured to power to drive

100% of thrust from the propeller section **14**. A scenario in which the electric machine **74** provides all power to the propeller section **14** can include shutdown of the engine **10**. In one non-limiting example of an engine being shut down, upon or near landing the engine **10** can be commanded to shut down and the electric machine **74** used to drive further propeller thrust requirements, whether that includes fine power on short-final or power when the propeller section **14** is configured in reverse pitch to aid in slowing the aircraft.

[0048] A battery or other secondary power source can be used to provide power to the electric machine **74** when operated as a motor. The supplementation of power by the electric machine **74** to the LP shaft **36** in this manner can be transitory or steady state, depending on the control requirements requested of the turboprop engine. For example, in those operating conditions in which power output of the engine lags behind a commanded power, the electric machine **74** can provide near instantaneous supplemental power to the LP shaft **36** to provide on-condition power output from the turboprop until the engine **10** achieves a steady state operating condition at the higher output power. In still further forms, the electric machine can be operated as a motor in some portions of operation of the engine **10**, and as a generator in other portions of operation of the engine **10**. along the lines of any of the variations discussed herein.

[0049] Given the proximity of the electric machine **74** to the intake channel **54**, the electric machine **74** can be further positioned to exchange heat with the inlet portion of the air **53** traversing the intake channel **54** to aid in removing heat from the electric machine **74**. Cooling of the electric machine **74** using the inlet portion of air **53** permits tighter packaging of the electric machine **74** and closer location of the electric machine **74** to heat generating portions of the engine **10**. The relative location of the electric machine **74** and intake channel **54** can permit an effective exchange of heat through any number of useful mechanisms including at least one of conduction and convection cooling. In some forms cooling air may directly impinge upon the electric machine **74** and/or may be used to vent a cavity in which the electric machine **74** is located. In still further forms the electric machine **74** can be used in part to form the flow path of the intake channel **54** to provide direct heat transfer between the electric machine **74** and the inlet portion of air **53**. In yet still further forms the electric machine **74** may be in direct contact with a portion of the intake channel **54** forming the flow path such that heat transfer occurs between the electric machine **74** and inlet portion of air **53** via that particular portion of the intake channel **54** forming the flow path. Further aspects of the location of electric machine **74** and various cooling techniques are described further below.

[0050] Various other embodiments are disclosed further herein related to the location and use of the electric machine **74**, the shape and configuration of the intake channel **54**, and various techniques to exchange heat between the electric machine **74** and the inlet portion of air **53** flowing in the intake channel **54**. As will be appreciated, like reference numerals refer to like elements and, thus, any of the variations disclosed herein related to any particular exemplary embodiment in any given figure are also applicable to embodiments depicted and discussed with respect to the other figures.

[0051] Turning now to FIG. 2, an exemplary embodiment of the turboprop engine **10** is illustrated in which the engine **10** includes an electric machine **74** located aft of the core turbine engine **16** in a tail cone **76** defining an aft end of an engine nacelle **78**. As will be appreciated, the engine nacelle **78** is used to enclose the gas turbine engine and includes one or more portions that interface with aircraft structure such as a wing, pylon, fuselage, etc. The tail cone **76** may be a complete body of revolution that circumferentially encloses the electric machine **74** in some embodiments. In alternative embodiments, however, the tail cone **76** may be a partial body of revolution or other shape that covers the electric machine to complete an enclosure with other aircraft structure (e.g., wing, pylon, fuselage, etc.). Thus, the tail cone **76** is any suitable structure of the engine nacelle which is located aft of the core turbine engine **16** and is used to wholly or partially enclose the electric machine **74**.

[0052] In some forms of the embodiments disclosed herein the engine nacelle **78** may take on the

form of an engine cowling when the engine **10** is installed on a single engine turboprop aircraft. In such installations the electric machine **74** can be located between the core turbine engine **16** and a firewall of the aircraft. In such installations, therefore, the engine cowling or other forebody structure of the aircraft can be used to enclose the electric machine **74**.

[0053] The embodiment of FIG. 2 includes an accessory gear box (AGB) **80** located aft of the HP compressor **22** and is coupled to a starter motor **82**. The starter motor **82** is coupled to the HP shaft via the AGB **80** such that during a start sequence of the turboprop engine **10** the starter motor **82** can be used to impart rotational power via the AGB **80** to the HP shaft **34**. In the illustrated embodiment, the AGB **80** is depicted as being co-axial with the HP shaft **34** and LP shaft **36** (it will be appreciated that although the AGB **80** is rotatably coupled to the HP shaft **34**, it is not otherwise rotatably coupled with the LP shaft **36**). To provide such a coaxial relationship between the HP shaft **34** and AGB **80**, in one form the AGB **80** is a planetary gear system in which the HP shaft **34** is coupled to a sun gear of the planetary gear system. In other forms a central gear of the AGB **80** is coupled via one or more idler gears to the starter motor **82**. Other forms are also contemplated to permit a co-axial relationship between the AGB **80** and HP shaft **34**. In other forms, however, the AGB **80** need not be co-axial with the HP shaft **34**.

[0054] The electric machine **74** is also depicted in FIG. 2 as being coupled to the LP shaft **36** through a speed change device **84** which can be used to alter a speed ratio between the LP shaft **36** and the electric machine **74**. In some forms, the speed change device **84** can be a transmission that provides a fixed speed ratio, but in other forms the transmission can provide variable speed ratios. In still other forms, the transmission can include a clutch mechanism to disengage the electric machine **74** from the LP shaft **36**. Still further, although the embodiment depicted in FIG. 2 includes the speed change device **84**, other embodiments need not include the device **84**. In such an embodiment, the electric machine **74** is directly connected to the LP shaft **36**.

[0055] The intake channel **54** forms a flow path between the AGB **80** and the HP compressor **22**. The flow path of the intake channel **54** follows the route depicted in FIG. 2, where the inlet portion of air **53** captured by the intake channel **54** passes first through a non-annular inlet **90** depicted at sightline A-A (an embodiment of which is illustrated in FIG. 7A, discussed in more detail below) before it is split into an annular flow path **92** depicted at sightline B-B (an embodiment of which is illustrated in FIG. 7B, discussed in more detail below). The non-annular inlet **90** is radially offset from the LP shaft **36** and confined to a circumferential section about the LP shaft **36**. The intake channel **54** takes the form of the annular shape leading up to the delivery of the inlet portion of air **53** to the HP compressor **22**. The intake channel **54** thus begins with a non-annular intake shape and ends with an annular shape. As will be appreciated, the annular shape of the intake channel includes a central interior that accommodates the LP shaft **36**. The intake channel **54** in the illustrated embodiment is thus required to morph, or change, from the non-annular shape at the inlet **90** to the annular shape at its discharge to the compressor **22** so that the intake channel **54** accommodates the intrusion of the LP shaft **36** through the intake channel **54** and to the electric machine **74**. In this way, the LP shaft **36** extends through at least a portion of the intake channel **54** where the flow path in the intake channel **54** changes shape to accommodate the LP shaft **36**. In some forms, an exterior surface of the LP shaft **36** may be exposed to the inlet portion of air **53** such that the inlet portion of air **53** flowing in the intake channel **54** is exposed to the rotating exterior surface of the LP shaft **36**. In other forms, however, the intake channel **54** may include a separate structure used to define the central interior and which is used to separate the exterior surface of the LP shaft **36** from the inlet portion of air **53** flowing in the intake channel **54**.

[0056] In some forms, the inlet **90** of the intake channel **54** is located at either a six o'clock position on the nacelle such as what would conventionally be considered the bottom, or underside, of the nacelle. An inlet on a Beechcraft Denali or Beechcraft King Air Turboprop are examples. The exhaust section **32** can be located at either or both of the three o'clock and nine o'clock position on the nacelle such as would conventionally be considered a left or right side of the nacelle. In this



manner, the spacing of the inlet **90** of the intake channel **54** is circumferentially displaced from the exhaust section **32** to minimize/prevent exhaust gases from being circulated to the inlet **90** for ingestion into the engine **10**. Furthermore, it will be appreciated that the inlet **52**, though illustrated at an axially aft location in the various embodiments, can be located forward closer to the blades **40** while still maintaining the configuration to reverse the flow from the direction of the incoming flow of air **50** to the axial flow direction required in the turbine engine configurations depicted. [0057] Also depicted in FIG. 2 is an offtake flow path **86** created by an offtake opening **88** provided in the intake channel **54** and which is configured to provide a flow of offtake air **94** to be used for heat exchange purposes with the electric machine **74**. The offtake opening **88** can be located downstream of the inlet **90** to the intake channel **54** and is structured to remove part of the inlet portion of air **53** flowing through the offtake channel **86**. The offtake opening **88** can be a permanent vent structure that includes a fixed opening through which air can pass regardless of mode of operation of the electric machine **74**. In other forms, however, the offtake opening **88** can include a movable mechanical structure that permits modulating the area of the offtake opening **88**, including in some forms fully closing the offtake opening. Such a movable mechanical structure can take any variety of forms such as a hinged plate, sleeve valve, or other suitable device. [0058] The offtake flow path **86** can take a variety of forms including the solid line depicted in FIG. 2 in which the flow path **86** flows past the electric machine **74**. In one form, the offtake flow path **86** can alternatively and/or additionally be routed along the dotted line shown in FIG. 2. Such a flow path can provide additional level of venting of the cavity in which the electric machine **74** is located, and/or provide greater dwell time within the cavity to ensure a higher level of heat exchange.

[0059] The offtake flow path can include one or more discharge openings **96** and **98** to permit the flow of offtake air **94** to exit from the nacelle **78**. The flow of offtake air **94** can be urged to exit through a pressure differential that exists between the offtake opening **88** and the discharge openings **96** and/or **98**. Such a pressure differential can be provide via ejector action if needed through suitable structure configured to provide such an action (e.g., a venturi ejector). One or both of the discharge openings **96** and **98**, in some embodiments, can be a permanent vent structure that includes a fixed opening through which air can pass regardless of mode of operation of the electric machine **74**.

[0060] In other forms, however, one or both of the discharge openings **96** and **98** can include a movable mechanical structure that permits modulating the area of the discharge openings **96** and **98**, including in some forms fully closing the discharge openings. Such a movable mechanical structure can take any variety of forms such as a hinged plate, sleeve valve, or other suitable device. The moveable mechanical structure can protrude into a passing flow of air to which the discharged flow of offtake air **94** is being discharged, and in other forms can protrude into the offtake flow path **86**.

[0061] Given the proximity of the electric machine **74** to the intake channel **54** and the configurations disclosed herein, various cooling techniques are contemplated with respect to the various embodiments. For example, though the passing flow of offtake air **94** is illustrated in FIG. 2 passing adjacent and/or around the electric machine **74**, in some embodiments, the passing flow of offtake air **94** can be directed to impinge directly upon a portion of the electric machine. Further, a surface of the electric machine **74**, such as an outer housing, can form part of the flow path of the offtake flow path **86**. In other forms, a structure forming the offtake flow path **86** can be in heat conductive relationship with a portion of the electric machine **74** (e.g., a housing of the electric machine **74**).

[0062] The embodiment depicted in FIG. 2 also includes a controller **100** configured to control various aspects of the depicted embodiment (the embodiment depicted in FIG. 1 can also include a controller for control of analogous features). As depicted through the various dotted lines, the controller **100** can control one or more different systems associated with operation of the engine **10**.

The dotted nature depicted in the figure denotes the optional inclusion of one or more, or all, of the systems connected with the controller **100**. For example, the controller **100** can be used to control variable stator vanes **70** in either or both turbines **28** and **30**. Additionally and/or alternatively, the controller **100** can be used to control variable stator vanes **60** in the compressor **22**. Additionally and/or alternatively, the controller **100** can be used to control fuel flow to the combustion section **26**. Additionally and/or alternatively, the controller **100** can be used to control one or more of the openings of the openings **88**, **96**, and **98**. Additionally and/or alternatively, the controller **100** can be used to control operation of the electric machine **74**. Additionally and/or alternatively, the controller **100** can command fuel flow to the combustion section **26** to be stopped and also simultaneously command the propeller section **14** to be positioned in a forward or reverse pitch configuration.

[0063] The controller **100** can thus be used in any or all of the following examples. The controller **100** may selectively drive the electric machine **74** as a generator. In those situations in which the electric machine **74** is 'powered on' to operate as a generator, the controller **100** can make adjustments to fuel rate delivery to the combustion section **26** while also optionally changing position of variable stator vanes **60** and/or **70**. The controller **100** can optionally operate a clutch in the speed change device **84**.

[0064] Turning now to FIG. **3**, another embodiment of the reverse flow engine **10** discussed above is illustrated. The exemplary engine **10** of FIG. **3** may be configured in substantially the same manner as the exemplary engine **10** of FIG. **2**, and as such the same or similar numbers may refer to the same or similar parts.

[0065] For the embodiment of FIG. **3**, the engine **10** includes an electric machine **74** coupled to the engine **10** and operated as a generator, the electric machine **74** further in electrical communication via a power conduit **102** with an electric machine **74b** which is operated as a motor. The electric machine **74b** operated as a motor is used to drive a set of propeller blades **40b** apart from the blades **40a** driven by the turbine engine **10** depicted at the top of FIG. **3**. The electric machine **74b** configured as a motor can be used to provide additional thrust output beyond that provided by the propeller blades **40a** driven by the turbine engine **10**.

[0066] Although the electric machine **74b** is illustrated apart from any turbomachinery components such as those at the top of the figure, in some embodiments the electric machine **74b** can be integrated with a gas turbine engine much in the same manner as the turboprop engine **10** depicted at the top of the figure. In these embodiments, the electrical coupling between the electric machines **74a** and **74b** can be used to exchange power between the two (e.g., where one machine is a motor and the other a generator) or can be coupled to a common energy storage device (e.g. a bank of batteries). Any of the variations in the embodiments discussed above are also applicable to the embodiments shown in FIG. **3**, such as but not limited to heat exchange between the electric machine **74** and the inlet portion of air **53**, the offtake flow path **86**, etc.

[0067] Turning now to FIG. **4**, another embodiment of the reverse flow engine **10** discussed above is illustrated. The exemplary engine **10** of FIG. **4** may be configured in substantially the same manner as the exemplary engine **10** of FIG. **2**, and, as such, the same or similar numbers may refer to the same or similar parts.

[0068] For the embodiment of FIG. **4**, the engine **10** includes an electric machine **74** coupled to the HP shaft **34** of the engine **10**, in which the configuration can employ the electric machine **74** as either a motor (e.g., to start the engine **10**) or a generator (e.g., to scavenge power for electric power generation). As will be appreciated in this embodiment, the LP shaft **36** need not be extended to the rear of the engine **10**.

[0069] Turning now to FIG. **5**, another embodiment of the reverse flow engine **10** discussed above is illustrated. The exemplary engine **10** of FIG. **5** may be configured in substantially the same manner as the exemplary engine **10** of FIG. **2**, and as such the same or similar numbers may refer to the same or similar parts.

[0070] For the embodiment of FIG. 5, the engine **10** includes electric machines **74a** and **74b**, each coupled with respective LP shaft **36** and HP shaft **34**. This embodiment enables power input/extraction to/from either spool **34,36** independently, as well as the potential to transfer power between the two spools **34,36** of the engine to improve operability/mitigate vibration, compressor stall or instability issues. As will therefore be appreciated, power can be extracted from spool **34** and provided to spool **36** in one mode of operation, power can be extracted from spool **36** and provided to spool **34** in another mode of operation, and power can be either extracted from or provided to both spools **34,36** in yet another mode of operation.

[0071] Any of the electric machines discussed herein are capable of being packaged so as to provide power densities suitable for use in the applications discussed herein. For example, the electric machines can have a power density ranging anywhere from greater than 3 k W/kg, greater than 5 kW/kg, and greater than 6 kW/kg.

[0072] As noted, the exemplary controller **100** useful in any of FIGS. 2-5 is configured to regulate any of the aforementioned systems such as variable stator vanes **60** and/or **70**, electric machine **74**, fuel flow to the combustion section **26**, etc., either alone or in combination, based on a control scheme stored in the controller **100**. In one or more exemplary embodiments, the controller **100** depicted in FIGS. 2 and 3 may be a stand-alone controller **100** for any of the aforementioned systems, or alternatively, may be integrated into one or more of a controller for the gas turbine engine with which the aforementioned systems are integrated, a controller for an aircraft including the gas turbine engine with which the aforementioned systems are integrated, etc.

[0073] Referring particularly to the operation of the controller **100**, in at least certain embodiments, the controller **100** can include one or more computing device(s) **104** such as depicted in FIG. 6. The computing device(s) **104** can include one or more processor(s) **104A** and one or more memory device(s) **104B**. The one or more processor(s) **104A** can include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, and/or other suitable processing device. The one or more memory device(s) **104B** can include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, RAM, ROM, hard drives, flash drives, and/or other memory devices.

[0074] The one or more memory device(s) **104B** can store information accessible by the one or more processor(s) **104A**, including computer-readable instructions **104C** that can be executed by the one or more processor(s) **104A**. The instructions **104C** can be any set of instructions that when executed by the one or more processor(s) **104A**, cause the one or more processor(s) **104A** to perform operations. In some embodiments, the instructions **104C** can be executed by the one or more processor(s) **104A** to cause the one or more processor(s) **104A** to perform operations, such as any of the operations and functions for which the controller **100** and/or the computing device(s) **104** are configured, the operations for any of the aforementioned systems such as variable stator vanes **60** and/or **70**, electric machine **74**, fuel flow to the combustion section **26**, etc., as described herein, and/or any other operations or functions of the one or more computing device(s) **104**. The instructions **104C** can be software written in any suitable programming language or can be implemented in hardware. Additionally, and/or alternatively, the instructions **104C** can be executed in logically and/or virtually separate threads on the one or more processor(s) **104A**. The one or more memory device(s) **104B** can further store data **104D** that can be accessed by the one or more processor(s) **104A**. For example, the data **104D** can include data indicative of power flows, data indicative of engine/aircraft operating conditions, and/or any other data and/or information described herein.

[0075] The computing device(s) **104** can also include a network interface **104E** used to communicate, for example, with the other components of system (e.g., via a communication network). The network interface **104E** can include any suitable components for interfacing with one or more network(s), including, for example, transmitters, receivers, ports, controllers, antennas, and/or other suitable components. One or more devices can be configured to receive one or more

commands from the computing device(s) **104** or provide one or more commands to the computing device(s) **104**.

[0076] The network interface **104E** can include any suitable components for interfacing with one or more network(s), including, for example, transmitters, receivers, ports, controllers, antennas, and/or other suitable components.

[0077] The technology discussed herein makes reference to computer-based systems and actions taken by and information sent to and from computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein can be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications can be implemented on a single system or distributed across multiple systems. Distributed components can operate sequentially or in parallel.

[0078] Turning now to FIGS. **7A** and **7B**, cross sectional views of two separate flow stations along the flow path of the intake channel **54** are illustrated. FIG. **7A** illustrates view A-A from FIGS. **2** and **3**, above, which depicts a cross section of the intake channel **54** that routes an inlet portion of air **53** of the flow of air **50**. The cross-section A-A depicted in FIG. **7A** is at or close to the inlet of the intake channel **54** and has a non-annular flow shape. The non-annular flow shape depicted in FIG. **7A** is similar to a kidney shape in the illustrated embodiment, but other non-annular shapes are also contemplated. The cross-section B-B depicted in FIG. **7B**, from FIGS. **2** and **3** above, is at or close to the discharge of the intake channel **54** as it delivers the inlet portion of air **53** of the flow of air **50** to the compressor **22**. The annular flow shape depicted in FIG. **7B** encloses the LP shaft **36**.

[0079] Referring now to FIG. **8**, a flow diagram of a method of operating a gas turbine engine in accordance with an exemplary aspect of the present disclosure is provided. The method of FIG. **7** may be utilized to operate one or more of the exemplary engine and electric machine placement described above with reference to FIGS. **1** through **5B**. Accordingly, it will be appreciated that the method may generally be utilized to operate a gas turbine engine including engine **10**, electric machine **74**, and intake channel **54**. However, in other exemplary aspects, the method may additionally or alternatively be utilized to operate any other suitable gas turbine engine.

[0080] More specifically, FIG. **8** discloses a method **106** of operating a reverse flow gas turbine engine which includes at **108** operating a gas turbine engine having a reverse flow configuration. The engine can take the form of a turboprop as discussed above in various embodiments. Step **110** includes receiving an incoming flow of air into an intake channel of the gas turbine engine. The flow of air received in the intake channel is turned at step **112** from its initial flow direction into an axial flow direction of the gas turbine engine. Step **114** discloses cooling an electric machine located aft of a turbine of the gas turbine engine using the flow of air in the intake channel.

[0081] The arrangement of the electric machine **74** coupled with the low pressure shaft **36** and located on a side of the high pressure compressor **22** opposite the high pressure turbine **28** provides various technical effects, including the ability to cool the electric machine **74** using the intake flow of air. Such a placement provides for additional separation from hot section components of the engine **10**. The electric machine **74** can be placed in proximity to the intake channel **54** at an aft end of the engine **10** in a heat exchange relationship, where the intake flow of air can exchange heat through either or both of conduction and convection with the electric machine **74**. In some forms an offtake flow of air can be extracted from the intake flow of air for use in cooling the electric machine **74** in lieu of the intake flow of air for additional flexibility, such as selective cooling provided through activation of discharge openings **96,98**. Various other flexible arrangements can also be provided of the electric machine **74**, and specifically cooling of the electric machine. For example, an electric machine **74b** can be coupled with the low pressure shaft **36**, while another electric machine **74a** is coupled with the high pressure shaft **34**. Both of electric machines **74a** and

74b can be located on a side of the high pressure compressor **22** opposite the high pressure turbine **28**.

[0082] It will further be appreciated that, during operation, a turbine engine is exposed to high temperatures, high pressures, and high speeds. For example, the turbine engine will generate excessive heat via combustion in operation. An aircraft engine will utilize a cooling system to cool engine components that reach relatively high temperatures. An engine's structural hardware is designed to support internal flow passages, with internal holes and passages, to allow cooling fluid to access high temperature components. However, the engine's structural hardware lacks the ability to define and control a variety of temperature zones throughout various sections of the internal hardware, resulting in a high thermal gradient between internal components. Further, the high thermal gradients in an engine's hardware inhibits the reliability and life of a given component and, thus, the reliability and life of the aircraft engine.

[0083] In operation, an aircraft engine experiences relatively high amounts of vibration and noise. Excessive vibration between components of an engine's internal hardware results in considerable stress concentrations as well as accelerated part damage and reduction in engine life. In particular, associated (e.g., close, proximate, etc.) components of an engine's hardware are susceptible to such vibrations and subsequent damage.

[0084] To address some of the issues presented by known engine casings and internal hardware, examples disclosed herein provide partition damper seal configurations for segmented internal cooling hardware. In certain examples, an engine structure (e.g., an engine casing, passage, liner, etc.) includes an inner wall (e.g., inner band) and an outer wall (e.g., outer band), the outer wall spaced apart from the inner wall in a radial direction. In certain examples, a space between the inner and outer walls defines a flow passage (e.g., cooling fluid cavity). In certain examples, a body (e.g., wire, damping mechanism, rib, etc.) is positioned between the inner and outer walls. In certain examples, the body extends in a radial direction and traverses the inner and outer walls. In certain examples, the body is detachably coupled to an inner surface of the outer wall and an outer surface of the inner wall. In certain examples, the body is integral to the engine structure. In certain examples, the body detaches from the inner and outer surfaces to at least partially seal the flow passage. In certain examples, the body detaches in response to the outer wall moving relative to the inner wall (e.g., vibration, thermal expansion, etc.).

[0085] In some examples, a first notch is disposed on the inner surface and opposes a second notch on the outer surface. Additionally or alternatively, the body is positioned between the notches. In some examples, the body contacts the first and second notches to seal the flow passage. In some examples, the body is a rib disposed on the inner surface or the outer surface. In some examples, the rib is positioned on a surface opposing a second rib and a third rib, wherein the second or the third rib contact the body to seal the flow passage. In some examples, the body is a damping mechanism between the inner and outer walls.

[0086] In some examples, the body is a three-dimensional (3D) printed wire. Accordingly, the 3D printed wire is integral to the inner and outer surfaces of the walls. In some examples, a plurality of bodies is arranged and distributed between the inner and outer walls. In some examples, the plurality of bodies detaches from the inner and outer surfaces in response to the inner wall moving relative to the outer wall. In some examples, the plurality of bodies seals the flow passage.

[0087] In some examples, the flow passage includes apertures extending through the inner and outer walls. Accordingly, the flow passage can be implemented as a cooling system for the engine structure. In some examples, the inner wall moves relative to the outer wall due to the thermal expansion of the engine structure (e.g., during engine operation).

[0088] Example partition damper seal configurations disclosed herein can be applied to closed and open rotor engine designs.

[0089] FIG. **9** is a cross-sectional view of an example sealing configuration **200**. An example sealing configuration **200** includes an inner wall **202** and an outer wall **204**. In some examples, the

inner wall **202** and the outer wall **204** define an engine structure (e.g., engine casing, turbine shroud, combustor liner, rotor passage, etc.). The inner wall **202** and the outer wall **204** are spaced apart in a radial direction. In the example as depicted in FIG. **2**, a body **206** is positioned between the inner wall **202** and the outer wall **204**. The example body **206** is disposed on an outer surface **208** of the inner wall **202** and an inner surface **210** of the outer wall **204**. Additionally or alternatively, the body **206** connects to the inner wall **202** and the outer wall **204** via a ligament **212** and a ligament **214**, respectively.

[0090] In this example, the space between the inner wall **202** and the outer wall **204** defines a flow passage **216**. The flow passage **216** has a first aperture **220** extending through the inner wall **202** and a second aperture **218** extending through the outer wall **204**. In some examples, the apertures **218**, **220** permit fluid flow to regions of an engine's internal hardware via the flow passage **216**, in directions as generally indicated by arrows **222**.

[0091] In the illustrated example, the outer wall **204** moves relative to the inner wall **202**. For example, the inner wall **202** can thermally deflect in a direction as generally indicated by an arrow **228** and the outer wall **204** can thermally deflect in a direction as generally indicated by an arrow **226**. The amount of the thermal expansion of the outer wall **204** is greater than the amount of the thermal expansion of the inner wall **202**. Thus, the deflection of the outer wall **204** results in a second relative position of the outer wall **204** as depicted by outlines **230**.

[0092] In the illustrated example of FIG. **9**, the example body **206** is detachably coupled to the inner wall **202** and the outer wall **204** via the ligaments **212** and **214**. In FIG. **9**, the ligaments **212** and **214** are detachable (e.g., removable) connections between the body **206** and the walls **202**, **204**. In response to the outer wall **204** moving relative to the inner wall **202**, a stress point **232** can occur at the ligament **212** such that the body **206** detaches from at least one of the outer wall **204** or the inner wall **202**. The example outlines **230** depict the movement of the body **206** from a first location to a second relative location. In this example, the body **206** remains attached to the outer wall **204**. In other examples, the body **206** detaches from both walls **202** and **204**.

[0093] In the illustrated example, the surfaces **210** and **208** have a notch **234** and a notch **236**, respectively, that extend into the flow passage **216**. The body **206** is positioned between the notches **234** and **236**. In response to the outer wall **204** moving relative to the inner wall **202**, the body **206** contacts the notches **234** and **236** to at least partially seal (e.g., partially seal, fully seal, partially close, close, etc.) the flow passage **216**.

[0094] Additionally or alternatively, the example body **206** can dampen (e.g., resist) the movement of the walls **202** and **204**. For example, relatively high external forces (e.g., vibration, pressure, torque, etc.) within an example airfoil can exert variable force on at least one of the walls **202** or **204**. In the example of FIG. **9**, the body **206** can absorb (e.g., mitigate, dampen, etc.) forces exerted on the walls **202** and **204**. Accordingly, the example body **206** can prevent damage of the walls **202** and **204** from external forces.

[0095] FIG. **10** illustrates an isometric view of an example strut fairing **300** in which examples disclosed herein can be implemented. The example strut fairing **300** can be mounted in an engine casing. In this example, view A is a location of an example double walled airfoil of the strut fairing **300** that will be described in further detail in FIG. **11**.

[0096] FIGS. **11** and **12** are cross-sectional views of the example strut fairing **300** of FIG. **10**. FIG. **11** illustrates a partial view of the strut fairing **300**. In particular, FIG. **11** illustrates the strut fairing **300** at view A of FIG. **10**.

[0097] In the illustrated example of FIG. **11**, the strut fairing **300** has example sealing configurations **402**. The example sealing configurations **402** of FIG. **11** are similar to the example sealing configuration **200** of FIG. **2**, but, instead, includes a plurality of sealing configurations **402** distributed throughout the strut fairing **300**. In some examples, the example sealing configuration **200** can be implemented as the example sealing configuration **402**. The sealing configurations **402** have bodies **206** positioned between the walls **202** and **204** and traverse the walls **202** and **204** in an

axial direction, as generally indicated by lines **404**.

[0098] The example strut fairing **300** includes a plurality of bodies **206** distributed between the inner wall **202** and the outer wall **204**. Accordingly, the example sealing configurations **402** can at least partially seal the flow passage **216** in response to the outer wall **204** moving relative to (e.g., thermally expanding relative to) the inner wall **202**.

[0099] FIG. **12** is another cross-sectional view of the example strut fairing **300** of FIG. **3**. In illustrated example of FIG. **12**, the inflow of fluid (e.g., cooling fluid, air, etc.) moves into the flow passage **216** in a direction as generally indicated by arrows **500**. In some examples, a pressurized cavity **502** pushes the fluid into the flow passage **216** corresponding to the direction of the arrows **500**. Fluid moves out of the flow passage **216** in a direction as generally indicated by arrows **504**.

[0100] The arrangement of the example sealing configurations **402** define zones **506**, **508**, **510**, **512**, and **514** (e.g., temperature zones) of the strut fairing **300**. In some examples, at least one of the zones **506**, **508**, **510**, **512**, and **514** may need to be a substantially different temperature (e.g., substantially higher temperature, substantially lower temperature, etc.) based on a design or an application of the strut fairing **300**.

[0101] FIG. **13** is an example engine structure **600** (e.g., an airfoil) that can be implemented in accordance with the teachings of the present disclosure. The example airfoil **600** of FIG. **13** is similar to the strut fairing **300** of FIGS. **10**, **11**, and **12**, but, instead, includes a plurality of sealing configurations **402** distributed in groups **602**, **604**, **606**, **608**, **610**, **612**, **614**, and **616** throughout the inner wall **202** and the outer wall **204**. In some examples, groups **602**, **604**, **606**, **608**, **610**, **612**, **614**, and **616** can provide multiple sealing and damping functions for the example airfoil **600**. For example, group **602**, with multiple sealing configurations **402**, can provide a greater degree of sealing and/or damping relative to group **616**, with a singular sealing configuration **402**.

[0102] FIG. **14** is a cross-sectional view of an example sealing configuration **700** that can be implemented in accordance with the teachings of the present disclosure. The example sealing configuration **700** of FIG. **14** is similar to the example sealing configuration **200** of FIG. **2**, but, instead, includes a body **206** with a first material **702**, and a second material **704**. The second material **704** is that of the walls **202** and **204**, and the first material **702** is that of the body **206**. In operation, the movement of the walls **202** and **204** can result in a stress concentration at point **706** due to a connection between first material **702** and the second material **704**.

[0103] FIG. **15** is a cross-sectional view of an example sealing configuration **800** that can be implemented in accordance with the teachings of the present disclosure. The example sealing configuration **800** of FIG. **15** is similar to the example sealing configuration **700** of FIG. **14**, but, instead, includes a body **206** with a third material **802**. The third material **802** can be a flexible (e.g., relatively soft, damping, etc.) core of the body **206**. In some examples, the first material **702** can be a wear resistant layer of the body **206**. In operation, the movement of the walls **202** and **204** can result in a relatively high stress concentration at point **804** due to a connection between first material **702** and the second material **704**.

[0104] FIG. **16** is a cross-sectional view of an example shroud hanger sealing configuration **900** that can be implemented in accordance with the teachings of the present disclosure. In some examples, the shroud hanger sealing configuration **900** can be included in an engine casing assembly. The example shroud hanger sealing configuration **900** of FIG. **16** is similar to the example sealing configuration **200** of FIG. **9**, but instead includes a first structure **902** having an outer wall **904** and a second structure **906** having an inner wall **908**. The body **206** is positioned between the outer wall **904** and the inner wall **908**. In some examples, the first structure **902** can be a wall of an engine's hardware (e.g., shroud wall, hanger wall, casing wall, a flow passage wall, etc.) that is different from the second structure **906**. Accordingly, the body **206** can be at least one of a sealing mechanism or a damping mechanism for structures **902**, **906** (e.g., casing, fairing, shroud, hanger, etc.) within an engine's hardware.

[0105] FIG. **17** is a cross-sectional view of an example sealing configuration **1000** that can be

implemented in examples disclosed herein. The example sealing configuration **1000** of FIG. **17** is similar to the example sealing configuration **200** of FIG. **9**, but, instead, includes ribs **1002** and **1004** disposed on the inner surface **210** of the outer wall **204** and a rib **1006** disposed (e.g., positioned) on the outer surface **208** of the inner wall **202**. The rib **1006** is positioned between the ribs **1002** and **1004**. In response to the inner wall **202** moving relative to the outer wall **204**, the rib **1006** contacts the ribs **1002** and **1004** to at least partially seal (e.g., fully seal, partially close, close, etc.) the flow passage **216**, as depicted by outlines **1008**.

[0106] Additionally or alternatively, the example rib **1006** can dampen (e.g., resist) the movement of the walls **202** and **204**. Example forces, exerted in a direction as generally indicated by arrows **1010** on at least one of the walls **202** or **204**, can result in a stress concentration point **1012**. Accordingly, the example rib **1006** can support (e.g., counter act the force on) the walls **202** and **204** from overpressure due to external forces.

[0107] FIG. **18** is a cross-sectional view of an example casing **1100** (e.g., a double wall casing) that can be implemented in examples disclosed herein. In some examples, the casing **1100** can include the airfoils and/or the sealing configurations described in FIG. **10**. In general, the example casing **1100** can have a flow passage positioned between the casing walls or any other double wall structure (e.g., airfoil) described in current disclosure. In particular, the flow between casing walls **202**, **204** can be used to cool and/or heat the casing **1100** or to deliver the cooling/heating flow elsewhere. The example casing **1100** has example sealing configurations **1102**. The example casing **1100** of FIG. **18** is similar to the example strut fairing **300** of FIGS. **10**, **11**, and **12**, but, instead, the casing **1100** includes multiple sealing configurations **1102** distributed (e.g., dispersed) throughout the inner wall **202** and the outer wall **204** of the casing **1100** (e.g., cooled casing).

[0108] FIG. **19** is a cross-sectional view of the example casing **1100** of FIG. **18** taken along a plane B-B as shown in FIG. **18**. In this example, the casing **1100** is depicted with a plurality of bodies **206** traversing the walls **202** and **204** in a first direction that is substantially perpendicular to the walls **202** and **204**.

[0109] FIG. **20** is a cross-sectional view of the example casing **1100** of FIG. **18** taken along a plane C-C as shown in FIG. **18**. In this example, the casing **1100** is depicted with the body **206** extending in a second direction that follows a curve of the walls **202** and **204**, as generally indicated by an arrow **1300**.

[0110] FIGS. **21A-21B** are top views of the casing **1100** as shown in FIG. **18**. In the example casing **1100** of FIGS. **18-21B**, the plurality of bodies **206** are arranged in a pattern (e.g., grid pattern, array, etc.) **1400** as illustrated in FIGS. **21A** and **21B**. FIG. **21A** is a perspective view of the pattern **1400**. FIG. **21B** is a detailed view of view D of the pattern **1400** as shown in FIG. **21A**. The bodies **206** extending in a first direction, as shown in FIG. **19**, intersect the bodies **206** extending in a second direction, as shown in FIG. **20**.

[0111] The examples of FIGS. **11-21B** illustrate various engine parts in which partition damper seal configurations can be implemented in accordance with the teachings of the present disclosure. Accordingly, the various engine parts illustrated in FIGS. **11-21B** utilize sealing configurations to provide for a sealing function and/or a damping function for segmented internal hardware of an aircraft (e.g., an aircraft engine, aircraft engine casing, etc.). In the example engine parts of FIGS. **11-21B**, a sealing configuration, a damper, and structures associated therewith provide sealing of a flow passage in the engine's hardware.

[0112] Although each example engine part disclosed above has certain features, it should be understood that it is not necessary for a particular feature of one example airfoil to be used exclusively with that example. Additionally or alternatively, although each example sealing configuration disclosed above has certain features, it should be understood that it is not necessary for a particular feature of one example sealing configuration to be used exclusively with that example. Instead, any of the features described above and/or depicted in the drawings can be combined with any of the examples, in addition to or in substitution for any of the other features of



those examples. One example's features are not mutually exclusive to another example's features. Instead, the scope of this disclosure encompasses any combination of any of the features.

[0113] In some examples, the example strut fairing **300**, the example airfoil **600**, or casing **1100** includes means for sealing the flow passage, such as the flow passage **216**. For example, the means for sealing (e.g., the sealing means) may be implemented by example sealing configurations **200**, **402**, **700**, **800**, **900**, **1000**, **1102**. In some examples, the means for sealing may be implemented by bodies **206** and/or the ribs **1002**, **1004**, **1006**.

[0114] In some examples, at least one of the inner wall **202** or the inner wall **908** includes first wall means. In some examples, at least one of the outer wall **204** or the outer wall **904** includes second wall means.

[0115] From the foregoing, it will be appreciated that example systems, methods, apparatus, and articles of manufacture have been disclosed that provide partition damper seal configurations for segmented internal cooling hardware. In examples disclosed herein, a sealing configuration, a damper, and/or structures associated therewith provide sealing of a flow passage in the engine's hardware and provide a damping function to hold surfaces of a component intact. In examples disclosed herein, cooling of the hardware may enable use of higher temperatures in the engine cycle. In examples disclosed herein, lower grade materials can be utilized in the engine. Further, in examples disclosed herein, the seal can be printed integrally via additive manufacturing, reducing assembly costs and providing sealing in complex geometries. Accordingly, examples disclosed herein enable increased life of an engine, mitigate high vibrations, and reduce high thermal gradients between components of an engine's hardware.

[0116] Further aspects are provided by the subject matter of the following clauses:

[0117] An aircraft engine assembly comprising: a gas turbine engine having a high pressure compressor, a high pressure turbine, a high pressure shaft coupling the high pressure compressor with the high pressure turbine, a low pressure turbine, and a low pressure shaft coupled to the low pressure turbine, the high pressure turbine located forward of the high pressure compressor, and the low pressure turbine located on a forward end of the gas turbine engine; an intake channel of the gas turbine engine configured to receive an incoming flow of air and form an intake flow of air, the intake channel configured to turn the received incoming flow of air from an incoming flow direction to a first axial direction of the gas turbine engine, the incoming flow direction reverse of the first axial direction; and an electric machine coupled with the low pressure shaft and located at the aft end of the gas turbine engine proximate the intake channel, the electric machine in heat exchange communication with the intake flow of air such that the electric machine transfers heat to the incoming flow of air within the intake channel when the electric machine is operated.

[0118] The aircraft engine assembly of one or more of these clauses, which further includes a tail cone, and wherein the electric machine is located within the tail cone.

[0119] The aircraft engine assembly of one or more of these clauses, wherein the low pressure shaft is oriented to extend through the intake channel such that an axial portion of the low pressure shaft is surrounded by the incoming flow of air in the intake channel.

[0120] The aircraft engine assembly of one or more of these clauses, wherein the electric machine is positioned to receive an impingement of the intake flow of air.

[0121] The aircraft engine assembly of one or more of these clauses, wherein an offtake flow of air is extracted from the intake flow of air, the offtake flow of air forming a cooling flow of air routed to the electric machine.

[0122] The aircraft engine assembly of one or more of these clauses, wherein the low pressure turbine includes variable stator vanes, wherein the gas turbine engine includes an engine controller, and wherein the variable stator vanes of the low pressure turbine are controlled by the controller to change position when the electric machine changes from a first power level to a second power level.

[0123] The aircraft engine assembly of one or more of these clauses, which further includes a

propeller located on a forward end of the gas turbine engine and coupled via the low pressure shaft with the low pressure turbine, and which further includes a gearbox coupled between the low pressure shaft and the propeller.

[0124] The aircraft engine assembly of one or more of these clauses, which further includes a propeller located on a forward end of the gas turbine engine and coupled via the low pressure shaft with the low pressure turbine, and wherein the propeller is variable pitch propeller, and wherein the electric machine is coaxial with the low pressure shaft.

[0125] The aircraft engine assembly of one or more of these clauses, which further includes an engine nacelle enclosing the gas turbine engine, the electric machine located in an enclosed space aft of the high pressure compressor.

[0126] The aircraft engine assembly of one or more of these clauses, wherein the engine nacelle includes a discharge opening sized to permit discharge of the portion of the intake flow of air.

[0127] The aircraft engine assembly of one or more of these clauses, wherein the electric machine is in a conductive heat exchange communication with the intake flow of air.

[0128] The aircraft engine assembly of one or more of these clauses, which further includes an engine nacelle enclosing the gas turbine engine, the electric machine located in an enclosed space aft of the high pressure compressor.

[0129] The aircraft engine assembly of one or more of these clauses, wherein the engine nacelle includes a discharge opening sized to permit discharge of the portion of the intake flow of air.

[0130] The aircraft engine assembly of one or more of these clauses, wherein the electric machine is in a conductive heat exchange communication with the intake flow of air.

[0131] The aircraft engine assembly of one or more of these clauses, wherein the intake channel includes a first portion that directs air into a first radial side of the low pressure shaft, wherein the intake channel includes a second portion configured as annular in shape, and wherein the first portion is upstream of the second portion, the second portion directing the intake flow of air in the first axial flow direction toward the high pressure compressor.

[0132] An aircraft powerplant comprising: a gas turbine engine having a high pressure compressor and a high pressure turbine, the gas turbine engine further having a high pressure shaft coupling the high pressure compressor with the high pressure turbine, the gas turbine engine also having a first axial flow direction from the high pressure compressor to the high pressure turbine; a propeller coupled to a low pressure turbine of the gas turbine engine using a low pressure shaft, the low pressure shaft located coaxial with the high pressure shaft, the propeller configured to receive a free stream flow of air oriented in a freestream direction and impart work upon the free stream flow of air, the propeller located on an upstream side of the freestream direction from the high pressure turbine; an intake channel defining an intake flow of air in fluid communication with the gas turbine engine, the intake channel configured to reverse the intake flow of air initially flowing in the freestream direction to the first axial flow direction of the gas turbine engine; and an electric machine coupled to the low pressure shaft and located on an opposite side of the gas turbine engine from the propeller, the electric machine positioned to be cooled by a portion of the intake flow of air defined by the intake channel.

[0133] The aircraft powerplant of one or more of these clauses, which further includes an engine nacelle enclosing the gas turbine engine, the electric machine located in an enclosed space aft of the high pressure compressor.

[0134] The aircraft powerplant of one or more of these clauses, wherein the engine nacelle includes a discharge opening sized to permit discharge of the portion of the intake flow of air.

[0135] The aircraft powerplant of one or more of these clauses, wherein the electric machine is in a conductive heat exchange communication with the intake flow of air.

[0136] The aircraft powerplant of one or more of these clauses, wherein the intake channel includes a first portion that directs air into a first radial side of the low pressure shaft, wherein the intake channel includes a second portion configured as annular in shape, and wherein the first portion is

upstream of the second portion, the second portion directing air in the first axial flow direction toward the high pressure compressor.

[0137] The aircraft powerplant of one or more of these clauses, wherein the gas turbine engine includes an engine controller, and wherein the engine controller commands a change in fuel flow rate to a combustion section of the gas turbine engine when the electric machine changes from a first power level to a second power level.

[0138] The aircraft powerplant of one or more of these clauses, wherein the low pressure turbine is a free turbine.

[0139] The aircraft powerplant of one or more of these clauses, an engine exhaust configured to receive an exhaust flow in the first axial flow direction from the low pressure turbine and discharge the exhaust flow having a second axial direction component reverse of the first axial direction.

[0140] A method of cooling an electric machine comprising: operating a gas turbine engine having a high pressure compressor, a high pressure turbine, and a high pressure shaft, the high pressure compressor coupled to the high pressure turbine via the high pressure shaft, the gas turbine engine also including a low pressure turbine coupled via a low pressure shaft with a propeller; receiving an incoming flow of air in an incoming flow of air direction into an intake channel of the gas turbine engine, the incoming flow of air used in a combustion process of the gas turbine engine, the intake channel forming an intake channel flow path and defining an intake flow air; turning the intake flow of air using the intake channel from the incoming flow of air direction to an axial flow direction of the gas turbine engine; and cooling an electric machine located aft of the low pressure turbine using the intake flow of air flowing through the intake channel flow path.

[0141] The method of cooling an electric machine of one or more of these clauses, wherein the cooling includes impingement cooling the electric machine using the intake flow of air.

[0142] The method of cooling an electric machine of one or more of these clauses, wherein turning the flow includes changing shape of the flow from a non-annular shape to an annular shape.

[0143] The method of cooling an electric machine of one or more of these clauses, wherein the electric machine is located in a tail cone located aft of the gas turbine engine.

[0144] An aircraft engine assembly including: a gas turbine engine having a high pressure compressor, a high pressure turbine, a high pressure shaft coupling the high pressure compressor with the high pressure turbine, a low pressure turbine, and a low pressure shaft coupled to the low pressure turbine, the high pressure turbine located forward of the high pressure compressor, and the low pressure turbine located on a forward end of the gas turbine engine; a propeller located on a forward end of the gas turbine engine and coupled via the low pressure shaft with the low pressure turbine; an intake channel of the gas turbine engine configured to receive an incoming flow of air and form an intake flow of air, the intake channel configured to turn the received incoming flow of air from an incoming flow direction to a first axial direction of the gas turbine engine, the incoming flow direction reverse of the first axial direction; and an electric machine coupled with the low pressure shaft and located on a side of the high pressure compressor opposite of the high pressure turbine and proximate the intake channel, the electric machine in heat exchange communication with the intake flow of air such that the electric machine transfers heat to the incoming flow of air within the intake channel when the electric machine is operated, wherein at least one of the gas turbine engine, the propeller, the intake channel, or the electric machine includes an engine structure including: an outer wall having a first rib spaced apart from a second rib on a surface of the outer wall; and an inner wall surrounded by the outer wall, the inner wall having a third rib protruding from the inner wall towards the surface of the outer wall, an end of the third rib detachably coupled to a portion of the surface of the outer wall, the portion of the surface positioned between the first rib and the second rib, the third rib and at least one of the first rib or the second rib to at least partially seal a flow passage between the inner and outer walls.

[0145] The aircraft engine assembly of any preceding clause, wherein the engine structure includes at least one of a fairing, a strut, a casing, a passage, or an airfoil.

[0146] The aircraft engine assembly of any preceding clause, wherein the third rib separates the inner wall from the outer wall.

[0147] The aircraft engine assembly of any preceding clause, wherein ends of the first and second ribs are spaced apart from the inner wall.

[0148] The aircraft engine assembly of any preceding clause, wherein a side of the third rib contacts a side of the first rib to at least partially seal the flow passage, the side of the third rib facing the side of the first rib.

[0149] The aircraft engine assembly of any preceding clause, wherein the first, second, and third ribs traverse the inner and outer walls in an axial direction.

[0150] The aircraft engine assembly of any preceding clause, wherein the surface is a first surface, further including a second surface of the outer wall opposing the first surface, the outer wall including an aperture extending through the first and second surfaces.

[0151] The aircraft engine assembly of any preceding clause, wherein the aperture is a first aperture, wherein the inner wall includes third and fourth opposing surfaces, the inner wall including a second aperture extending through the third and fourth surfaces, the first, second, and third ribs positioned between the first and second apertures.

[0152] The aircraft engine assembly of any preceding clause, wherein the first aperture is adjacent to the first rib.

[0153] The aircraft engine assembly of any preceding clause, which further includes a tail cone, and wherein the electric machine is located within the tail cone.

[0154] The aircraft engine assembly of any preceding clause, wherein the low pressure shaft is oriented to extend through the intake channel such that an axial portion of the low pressure shaft is surrounded by the incoming flow of air in the intake channel.

[0155] The aircraft engine assembly of any preceding clause, wherein the electric machine is positioned to receive an impingement of the intake flow of air.

[0156] The aircraft engine assembly of any preceding clause, wherein an offtake flow of air is extracted from the intake flow of air, the offtake flow of air forming a cooling flow of air routed to the electric machine.

[0157] The aircraft engine assembly of any preceding clause, wherein the low pressure turbine includes variable stator vanes, wherein the gas turbine engine includes an engine controller, and wherein the variable stator vanes of the low pressure turbine are controlled by the engine controller to change position when the electric machine changes from a first power level to a second power level.

[0158] The aircraft engine assembly of any preceding clause, which further includes a propeller located on a forward end of the gas turbine engine and coupled via the low pressure shaft with the low pressure turbine, and which further includes a gearbox coupled between the low pressure shaft and the propeller.

[0159] The aircraft engine assembly of any preceding clause, which further includes a propeller located on a forward end of the gas turbine engine and coupled via the low pressure shaft with the low pressure turbine, and wherein the propeller is a variable pitch propeller, and wherein the electric machine is coaxial with the low pressure shaft.

[0160] An aircraft powerplant including: a gas turbine engine having a high pressure compressor and a high pressure turbine, the gas turbine engine further having a high pressure shaft coupling the high pressure compressor with the high pressure turbine, the gas turbine engine also having a first axial flow direction from the high pressure compressor to the high pressure turbine; a propeller coupled to a low pressure turbine of the gas turbine engine using a low pressure shaft, the low pressure shaft located coaxial with the high pressure shaft, the propeller configured to receive a free stream flow of air oriented in a freestream direction and impart work upon the free stream flow of air, the propeller located on an upstream side of the freestream direction from the high pressure turbine; an intake channel defining an intake flow of air in fluid communication with the gas

turbine engine, the intake channel configured to reverse the intake flow of air initially flowing in the freestream direction to the first axial flow direction of the gas turbine engine; an electric machine coupled to the low pressure shaft and located on an opposite side of the gas turbine engine from the propeller, the electric machine positioned to be cooled by a portion of the intake flow of air defined by the intake channel, wherein at least one of the gas turbine engine, the propeller, the intake channel, or the electric machine includes an engine structure including: an outer wall having a first rib spaced apart from a second rib on a surface of the outer wall; and an inner wall surrounded by the outer wall, the inner wall having a third rib protruding from the inner wall towards the surface of the outer wall, an end of the third rib detachably coupled to a portion of the surface of the outer wall, the portion of the surface positioned between the first rib and the second rib, the third rib and at least one of the first rib or the second rib to at least partially seal a flow passage between the inner and outer walls.

[0161] The aircraft powerplant of any preceding clause, wherein the engine structure includes at least one of a fairing, a strut, a casing, a passage, or an airfoil.

[0162] The aircraft powerplant of any preceding clause, wherein the low pressure shaft is oriented to extend through the intake channel such that an axial portion of the low pressure shaft is surrounded by the incoming flow of air in the intake channel.

[0163] The aircraft powerplant of any preceding clause, wherein the low pressure turbine includes variable stator vanes, wherein the gas turbine engine includes an engine controller, and wherein the variable stator vanes of the low pressure turbine are controlled by the engine controller to change position when the electric machine changes from a first power level to a second power level.

[0164] This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

## Claims

1. An aircraft engine assembly comprising: a gas turbine engine having a high pressure compressor, a high pressure turbine, a high pressure shaft coupling the high pressure compressor with the high pressure turbine, a low pressure turbine, and a low pressure shaft coupled to the low pressure turbine, the high pressure turbine located forward of the high pressure compressor, and the low pressure turbine located on a forward end of the gas turbine engine; a propeller located on the forward end of the gas turbine engine and coupled via the low pressure shaft with the low pressure turbine; an intake channel of the gas turbine engine configured to receive an incoming flow of air and form an intake flow of air, the intake channel configured to turn the received incoming flow of air from an incoming flow direction to a first axial direction of the gas turbine engine, the incoming flow direction reverse of the first axial direction; and an electric machine coupled with the low pressure shaft and located on a side of the high pressure compressor opposite of the high pressure turbine and proximate the intake channel, the electric machine in heat exchange communication with the intake flow of air such that the electric machine transfers heat to the incoming flow of air within the intake channel when the electric machine is operated, wherein at least one of the gas turbine engine, the propeller, the intake channel, or the electric machine includes a structure including: an inner wall and an outer wall, the outer wall spaced apart from the inner wall to define a flow passage between the outer wall and the inner wall; and a body positioned between the outer wall and the inner wall, the body detachably coupled to a surface of at least one of the outer wall or

the inner wall to at least partially seal the flow passage.

2. The aircraft engine assembly of claim 1, wherein the structure includes at least one of a fairing, a strut, a casing, a hanger, a shroud, or an airfoil.

3. The aircraft engine assembly of claim 1, wherein the body is a damper between the inner wall and the outer wall.

4. The aircraft engine assembly of claim 1, wherein the outer wall is movable with respect to the inner wall based on thermal expansion.

5. The aircraft engine assembly of claim 1, wherein the body is an additively manufactured seal.

6. The aircraft engine assembly of claim 5, wherein the additively manufactured seal is a wire.

7. The aircraft engine assembly of claim 1, wherein the body is detachably coupled to the surface of at least one of the outer wall or the inner wall with a ligament.

8. The aircraft engine assembly of claim 7, further including at least one aperture extending through at least one of the inner wall or the outer wall to the flow passage.

9. The aircraft engine assembly of claim 1, further including a plurality of bodies distributed between the inner wall and the outer wall.

10. The aircraft engine assembly of claim 1, which further includes a tail cone, and wherein the electric machine is located within the tail cone.

11. The aircraft engine assembly of claim 1, wherein the low pressure shaft is oriented to extend through the intake channel such that an axial portion of the low pressure shaft is surrounded by the incoming flow of air in the intake channel.

12. The aircraft engine assembly of claim 1, wherein the electric machine is positioned to receive an impingement of the intake flow of air.

13. The aircraft engine assembly of claim 1, wherein an offtake flow of air is extracted from the intake flow of air, the offtake flow of air forming a cooling flow of air routed to the electric machine.

14. The aircraft engine assembly of claim 1, wherein the low pressure turbine includes variable stator vanes, wherein the gas turbine engine includes an engine controller, and wherein the variable stator vanes of the low pressure turbine are controlled by the engine controller to change position when the electric machine changes from a first power level to a second power level.

15. The aircraft engine assembly of claim 1, further including a gearbox coupled between the low pressure shaft and the propeller.

16. The aircraft engine assembly of claim 1, wherein the propeller is a variable pitch propeller, and wherein the electric machine is coaxial with the low pressure shaft.

17. An aircraft powerplant comprising: a gas turbine engine having a high pressure compressor and a high pressure turbine, the gas turbine engine further having a high pressure shaft coupling the high pressure compressor with the high pressure turbine, the gas turbine engine also having a first axial flow direction from the high pressure compressor to the high pressure turbine; a propeller coupled to a low pressure turbine of the gas turbine engine using a low pressure shaft, the low pressure shaft located coaxial with the high pressure shaft, the propeller configured to receive a free stream flow of air oriented in a freestream direction and impart work upon the free stream flow of air, the propeller located on an upstream side of the freestream direction from the high pressure turbine; an intake channel defining an intake flow of air in fluid communication with the gas turbine engine, the intake channel configured to reverse the intake flow of air initially flowing in the freestream direction to the first axial flow direction of the gas turbine engine; an electric machine coupled to the low pressure shaft and located on an opposite side of the gas turbine engine from the propeller, the electric machine positioned to be cooled by a portion of the intake flow of air defined by the intake channel, wherein at least one of the gas turbine engine, the propeller, the intake channel, or the electric machine includes a structure including: an inner wall and an outer wall, the outer wall spaced apart from the inner wall to define a flow passage between the outer wall and the inner wall; and a body positioned between the outer wall and the inner wall, the body

detachably coupled to a surface of at least one of the outer wall or the inner wall to at least partially seal the flow passage.

**18.** The aircraft powerplant of claim 17, wherein the structure includes at least one of a fairing, a strut, a casing, a hanger, a shroud, or an airfoil.

**19.** The aircraft powerplant of claim 17, wherein the body is an additively manufactured wire.

**20.** The aircraft powerplant of claim 17, wherein the body is detachably coupled to the surface of the at least one of the outer wall or the inner wall with a ligament.

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